

Hybrid Intelligence

The next AI generation

AI that ~~replaces~~ **collaborates with** people

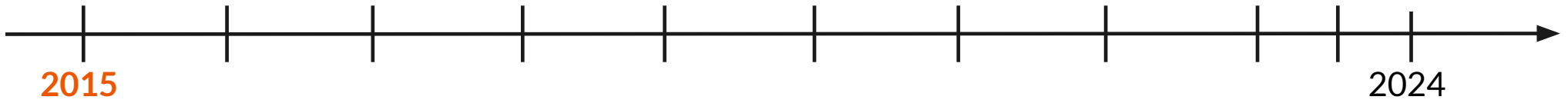


Logistics : Video Voting

- Strict deadline to vote : this Thursday **Oct 17th at 23:59**
- Vote the best video between **today and this Thursday**
- You need to vote 2 best videos (1 per pool)



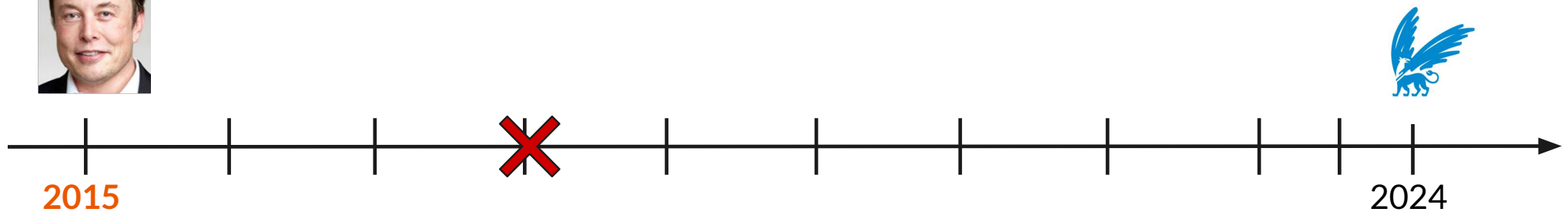
The current situation

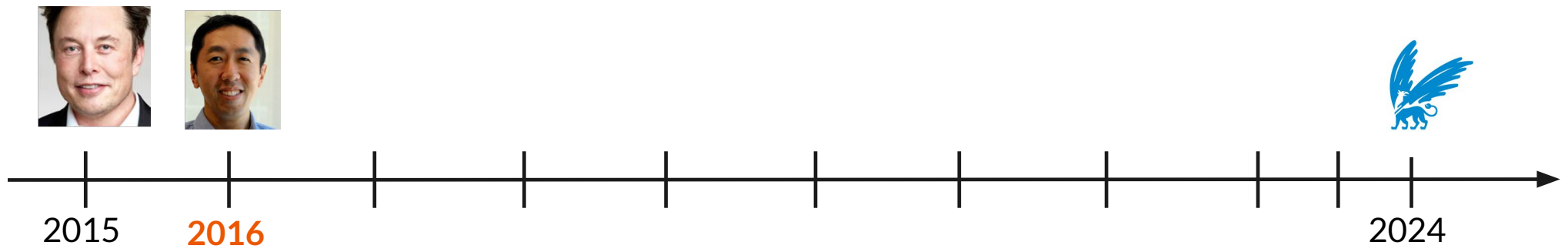




The current situation

“My guess for when we will have **full autonomy** in cars is approximately three years”
(Elon Musk, 2015)



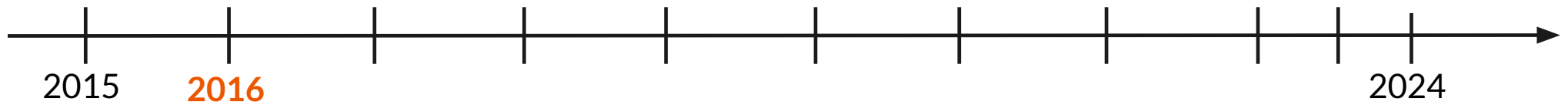




The current situation

“A highly-trained and specialised radiologist may now be in greater danger of **being replaced by a machine** than his own executive assistant”

(Andrew Ng, The Economist, 2016)

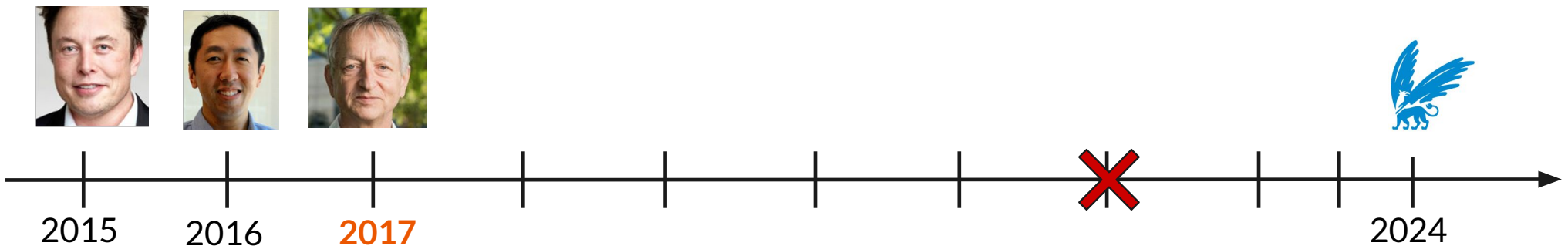




The current situation

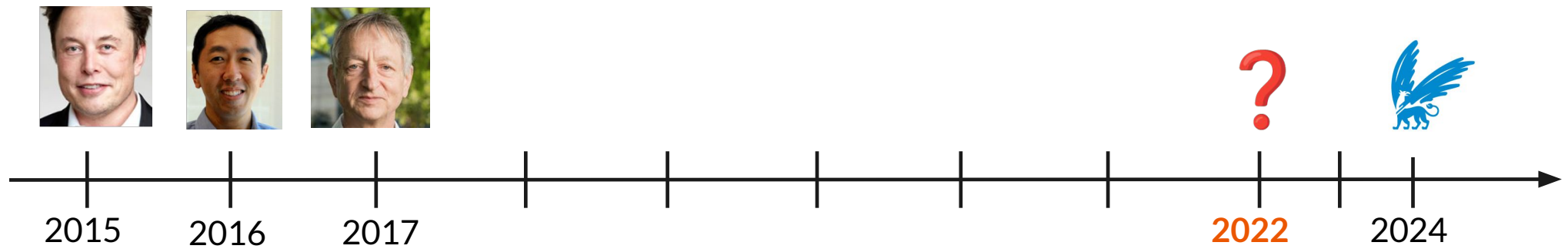
“People should stop training as radiologists now. It’s just completely obvious that within 5 years, **deep learning is going to do better** than radiologists”

(Geoffrey Hinton, The New Yorker, 2017)

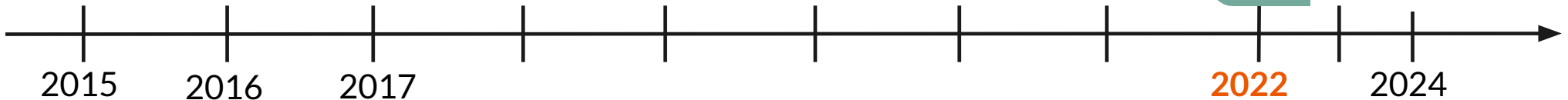




The current situation



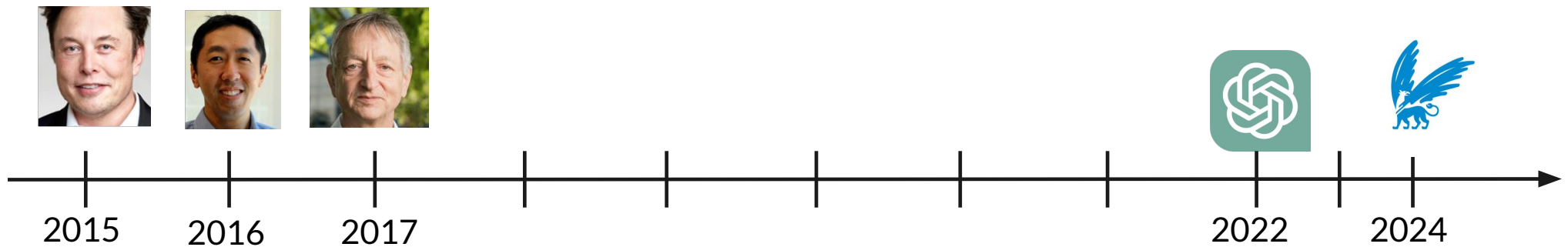
The current situation





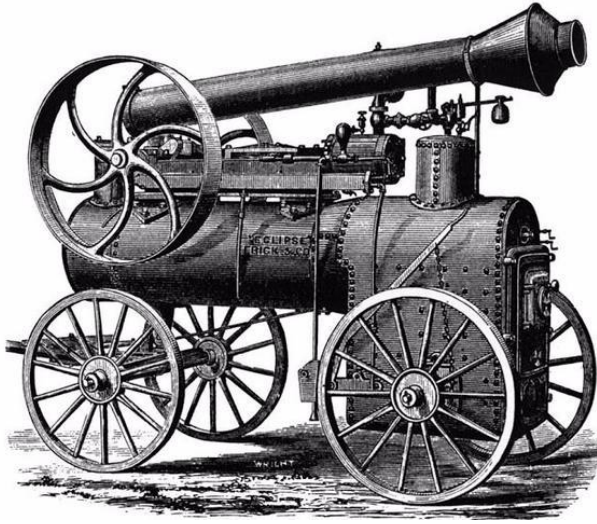
The current situation

Do we need to **replace** humans, and **why**?



Let's take a few steps back

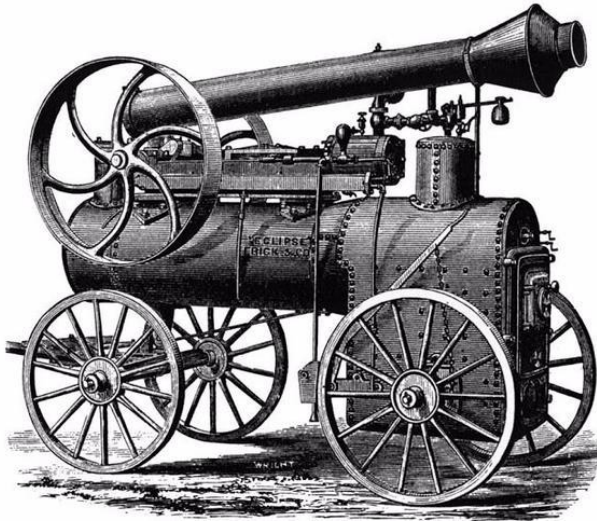
Science helped **extending** human capabilities



steam engine = **muscle power**

Let's take a few steps back

Science helped **extending** human capabilities



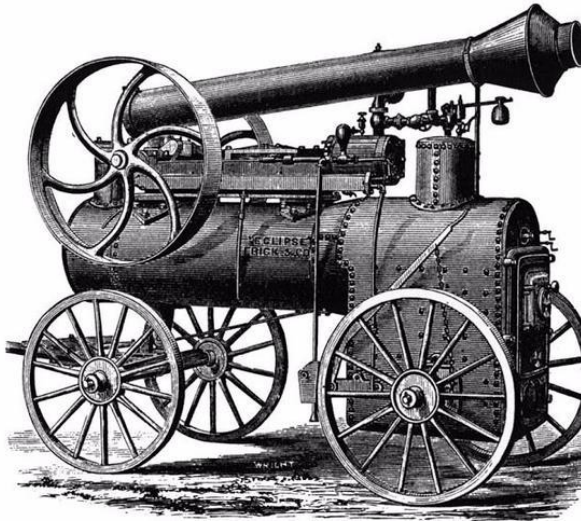
steam engine = **muscle power**



car = **human speed**

Let's take a few steps back

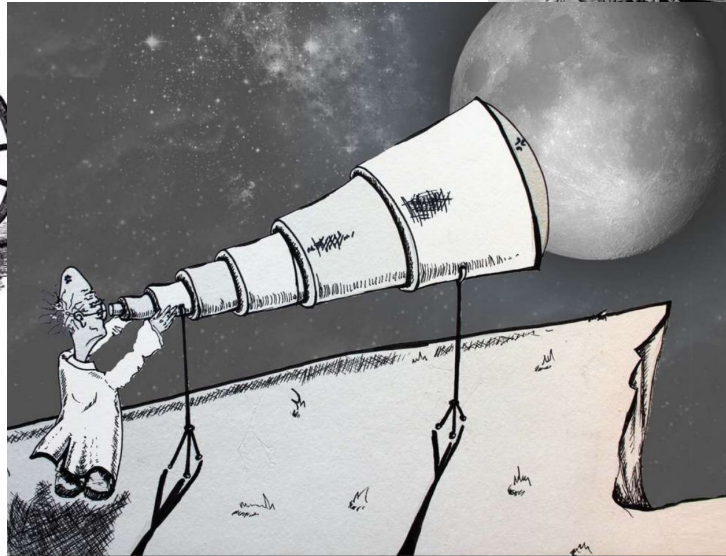
Science helped **extending** human capabilities



steam engine = **muscle power**



car = **human speed**



telescopes = **human vision**

AI should not imitate human intelligence



Imagine if planes were imitating birds.

AI should not replace human intelligence



It'd be like fitting the wrong nail

AI should collaborate with human intelligence



There comes Hybrid Intelligence!



Because AI and humans need each other

Humans have cognitive limitations

- Bad at probabilities
- Stereotypes
- Cognitive biases
- In-group favouritism
- Memory shortage

Machines have ability limitations

- Poor generalisation
- No/Little creativity
- Unawareness of societal values
- Cannot deal with rare/unique events



The Challenges ahead (“Problem Space”)

How to build hybrid intelligent systems that

- **augment** (rather than replace) human intelligence,
- leverage our strengths, and **compensate** for our weaknesses,
- **adapt** to situations, and
- take into account ethical, legal, and **societal considerations?**



HI Research Agenda

Where do we start?

4 functionalities

Collaboration

Adaptivity

Responsibility

Explainability

...and a theory for HI



HI Research Agenda

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Collaboration

Aadaptivity

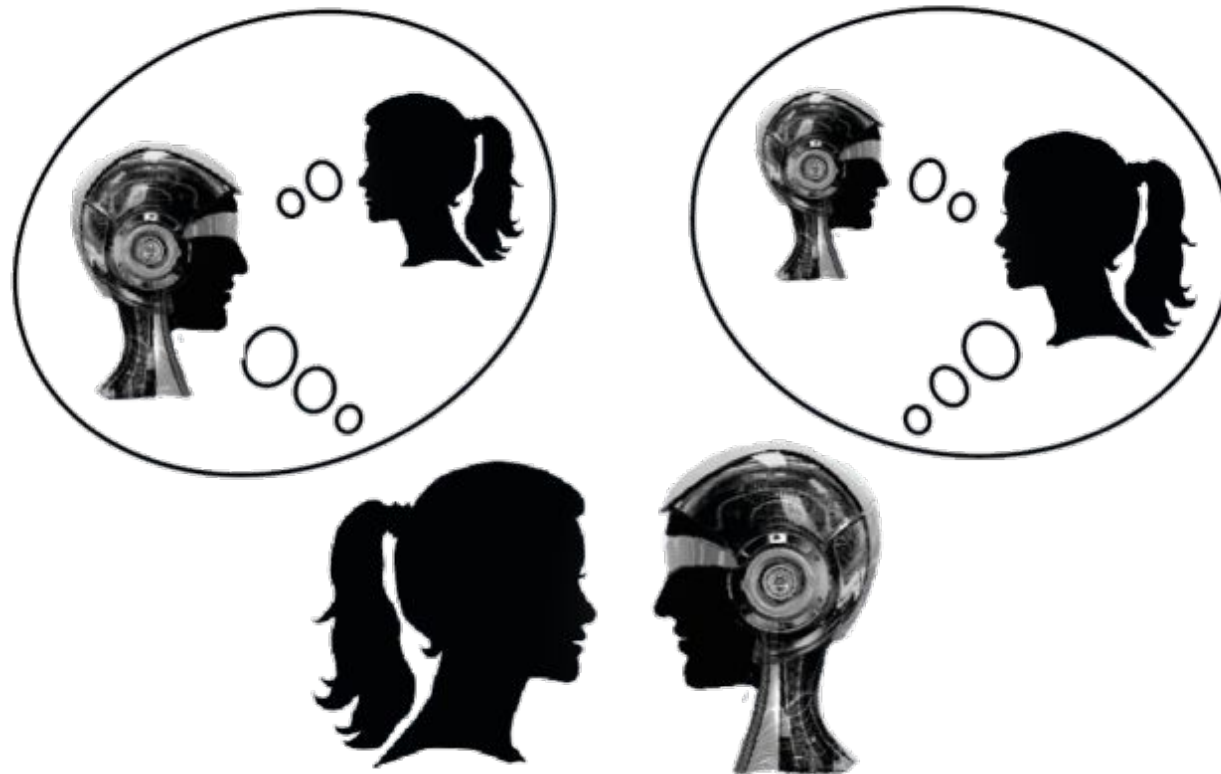
Responsibility

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1. Collaborative HI

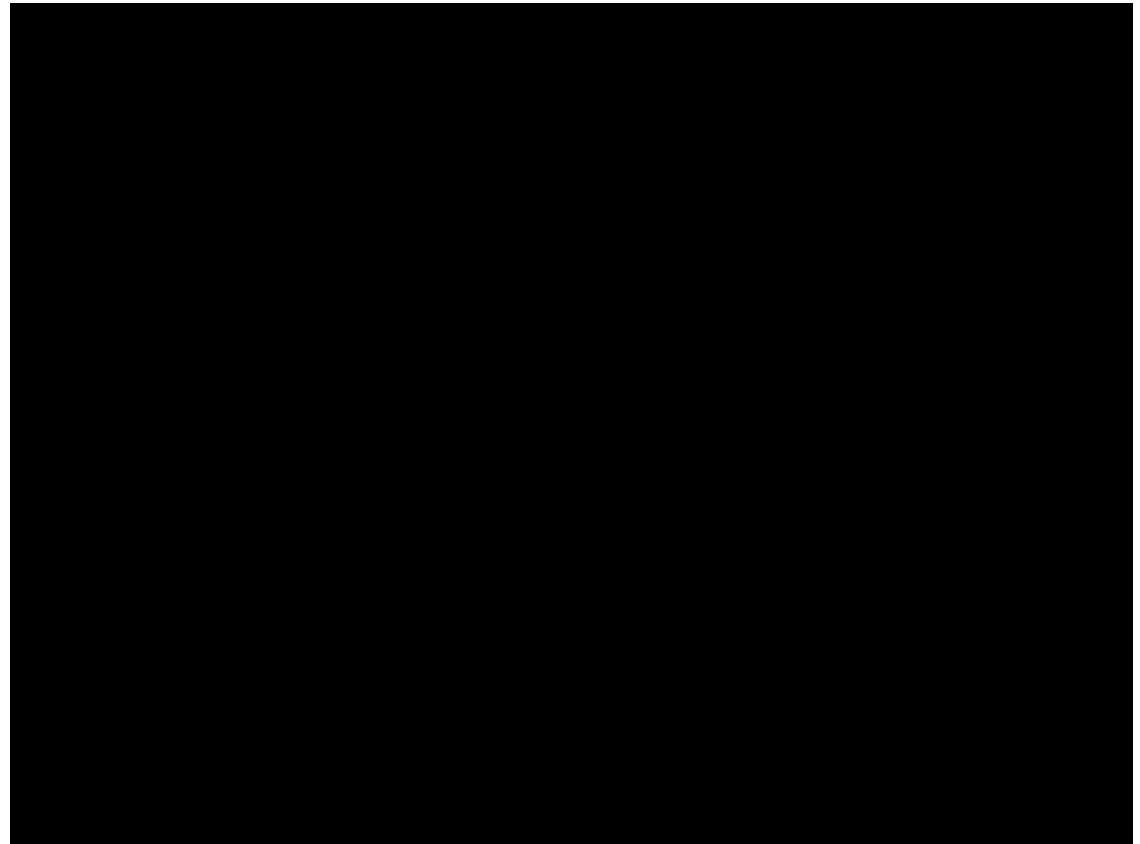
Humans do mindreading all the time





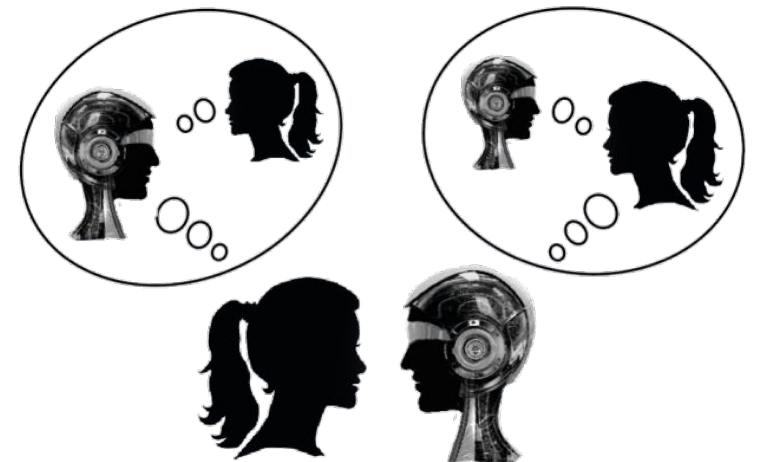
1. Collaboration

We start as children!



1. Collaboration

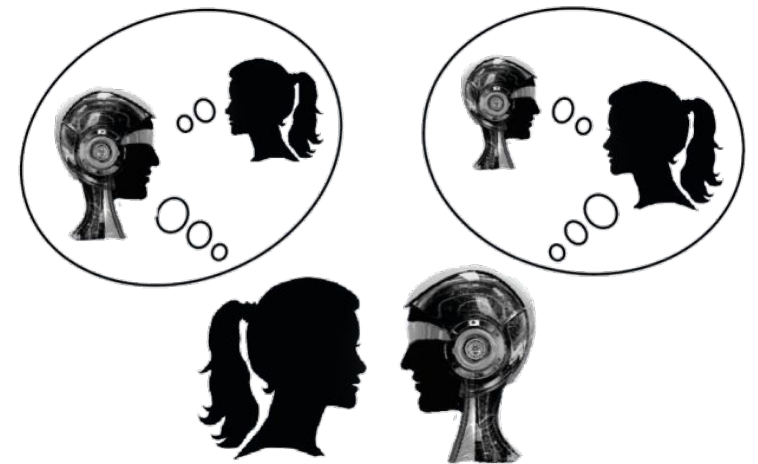
HI should have not only theory about the world, but a **theory** about **other agents' theories** of the world.



1. Collaboration

Theory of mind (ToM) ?

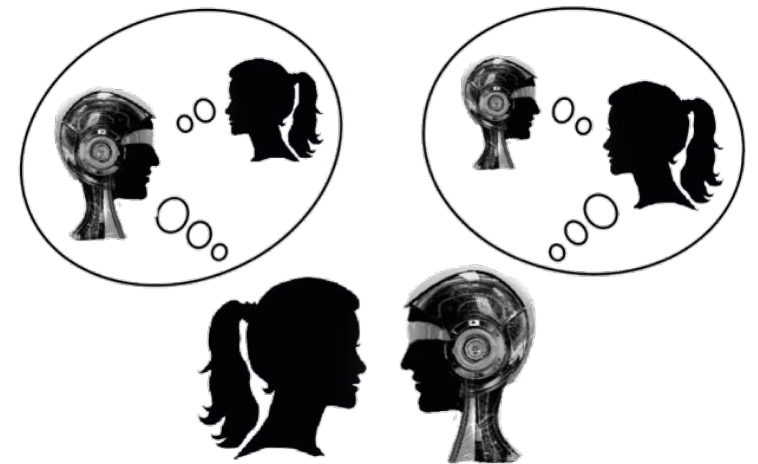
“The ability to attribute a wide range of mental states to the others, including intentions, hopes, expectations, imaginings, desires and beliefs, and thereby interpret, explain, and predict their behaviour.”



1. Collaboration

Theory of mind (ToM) ?

*“The ability to **attribute** a wide range of **mental states** to the others, including intentions, hopes, expectations, imaginings, desires and beliefs, and thereby **interpret, explain, and predict** their behaviour.”*



1. Collaboration

“2nd-order ToM is beneficial in **competitive**, **cooperative**, and mixed-motive situations.

Agents with deeper ToM levels give **better support** to humans on negotiation outcomes”

(de Weerd et al, AI Journal 2013)





1. Challenges for a Collaborative HI

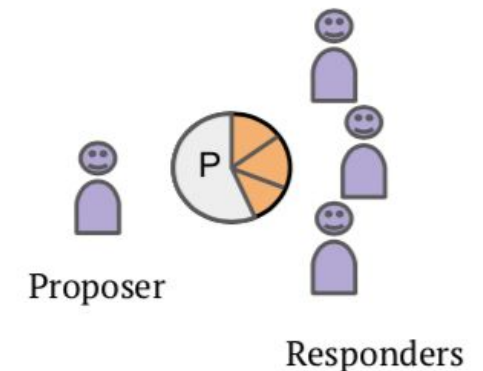
Beyond the “human-in-the-loop” : aim for **synergy and reciprocity**

- **Perceive** social behaviour by collaborators (through language and vision)
- **communicate** with their collaborators (dialogue, multimodal interaction...)
- understand **joint actions** in teams
- understand essential **social norms** for teamwork (reciprocity, hierarchy/leadership, in/out-group favouritism)



1. An example

- A dataset to **predict partner selection**
 - Issue from social psychology theories
 - dyadic social interactions
 - Multi-modality (audio-visual recordings)
- 300 participants
 - measures such as personality, trust, social-anxiety
 - **Cooperative and Collaborative tasks** (mostly involving money distribution)
- Goals
 - understand and predict interdependence in social interactions
 - Implement main social cues in **human-robot interaction** scenarios





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...and a **theory** for HI

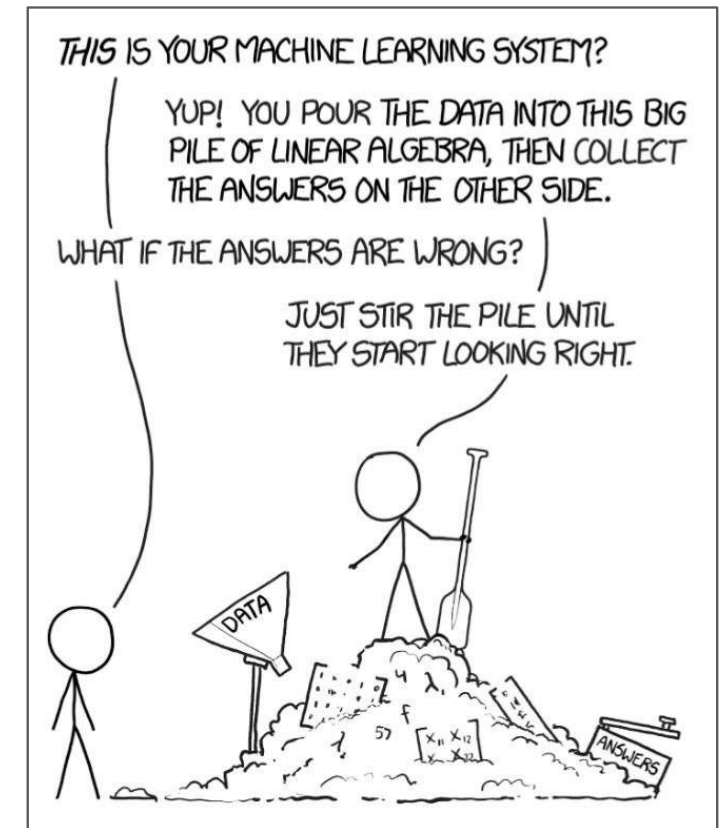
2. Adaptivity

Current ML cannot :

- adapt to changes in the **environment**
- adapt to changes in the **team**
- balance between **safety and reliability**

State of the art:

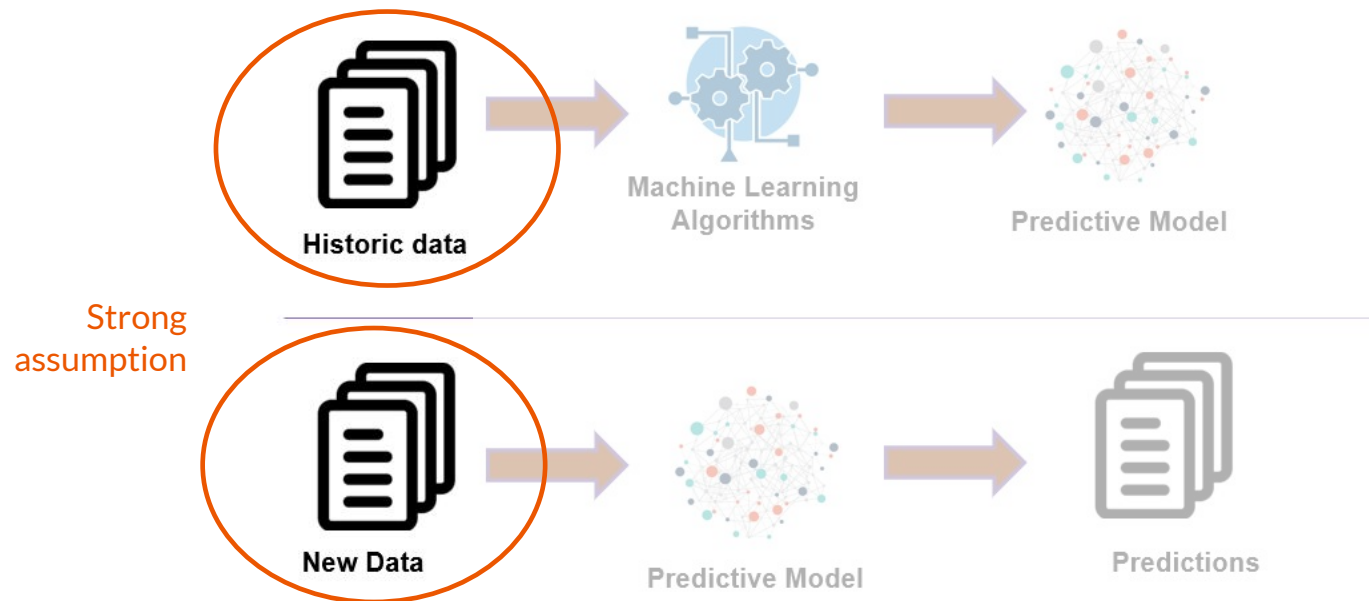
- Transfer learning
- Multi-task learning
- Meta-learning (learn from metadata about the experiments)



2. Adaptivity

Machine Learning

- Learn on historical data (**Training**)
- Apply over new data (**Predicting**) —> hardly similar in the real world !





Machine Bias

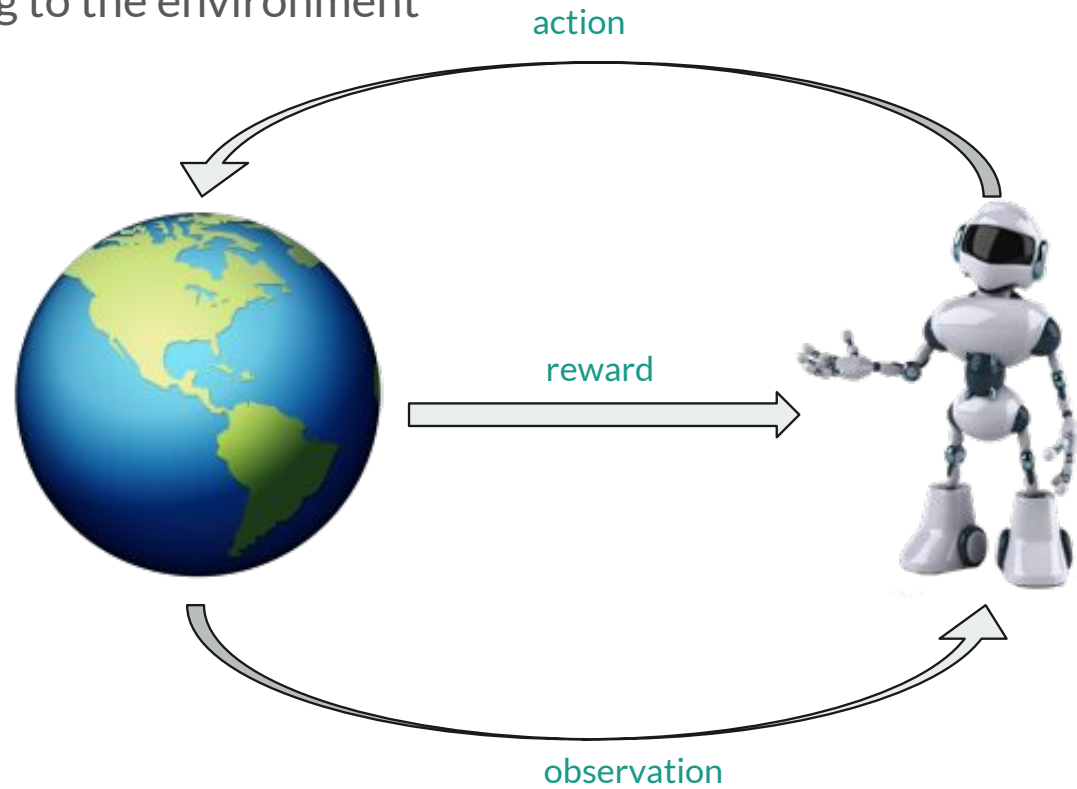
There's software used across the country to predict future criminals. And it's biased against blacks.

<https://www.propublica.org/article/machine-bias-risk-assessments-in-criminal-sentencing>

2. Adaptivity

Reinforcement Learning for the rescue

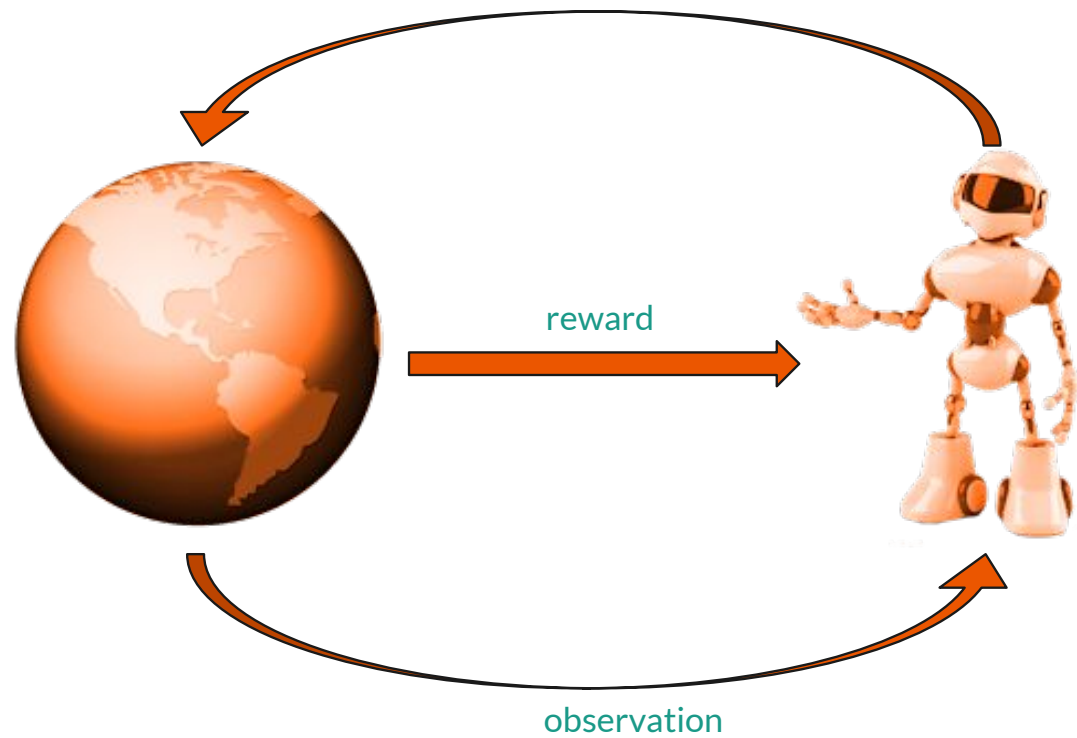
- i.e. adapting to the environment



2. Adaptivity

Reinforcement Learning for the rescue

- rewards&actions in different worlds action





2. Challenges for an Adaptive HI

Learn **reusable or safe** behaviours

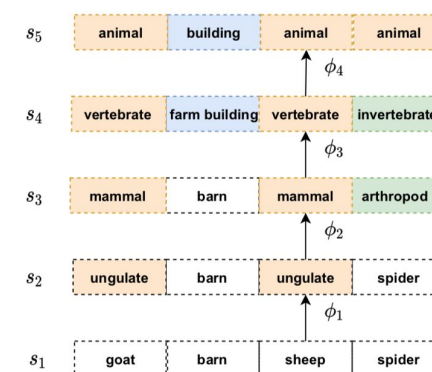
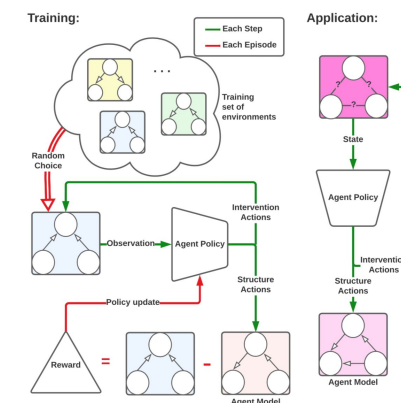
- in the reward/loss functions (*preferably don't do X*)
- as symbolic constraints (*never do X*)
- as restriction on the exploration process (*don't try X*)
- as preferences over the exploration process (*preferably try X*)

Learn **behaviours in fewer trials**

2. Examples

HI agents learning through interaction with the world

- Synthetic RL settings
- Use **shared representations** between humans and agents (causal models, KGs, vision LMs, etc)
- Goal :
 - **generalisation** over unseen data
 - **sample efficiency** (learning with less samples)



Sauter et al (2022). A Meta-Reinforcement Learning Algorithm for Causal Discovery. CLEAR2023

Höpner et al (2022.). Leveraging Class Abstraction for Commonsense Reinforcement Learning via Residual Policy Gradient Methods. IJCAI-22.



HI Research Agenda

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...and a **theory** for HI



3. Responsibility

AI is increasingly making key decisions

- for **individuals** (job selection, financial decisions, medical screening)
- for **society** (spam filtering, fake news & hate speech detection)

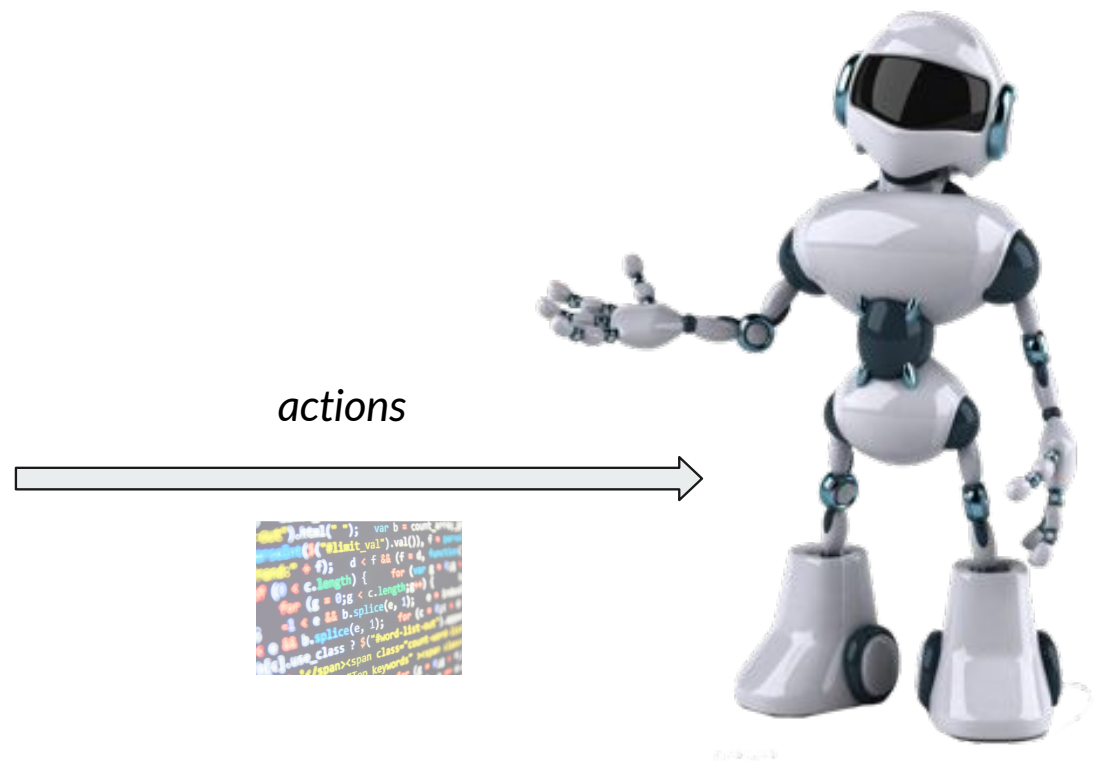
Reasons for these decisions?

- often unknown
- cannot be disputed
- **Explanations** need to be grounded in **values, norms, goals**

3. Responsibility

The AI approach : **implicit** human values

(e.g. the forbidden actions are not programmed, but the system/robot is not aware of why)



3. Responsibility

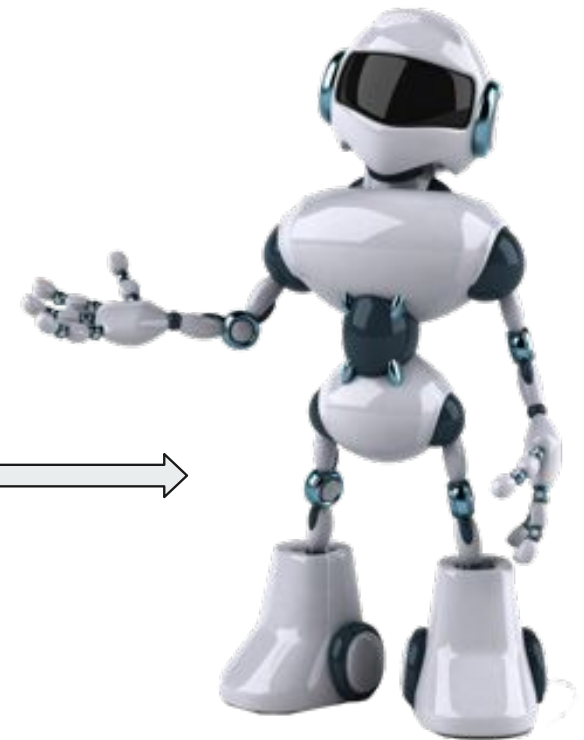
New approach : ethics **by design** (e.g. the system knows why values are forbidden)



values
ethics
norms



actions

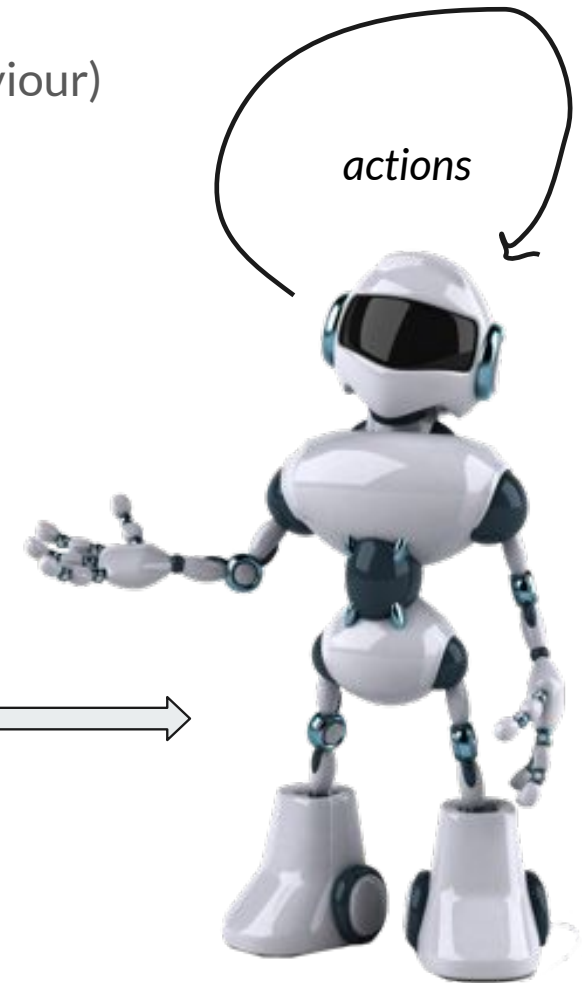


3. Responsibility

New approach : ethics **by design** (e.g. the system adapts his behaviour)



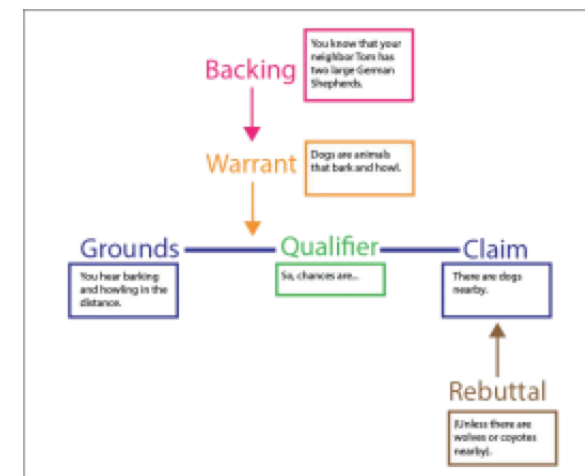
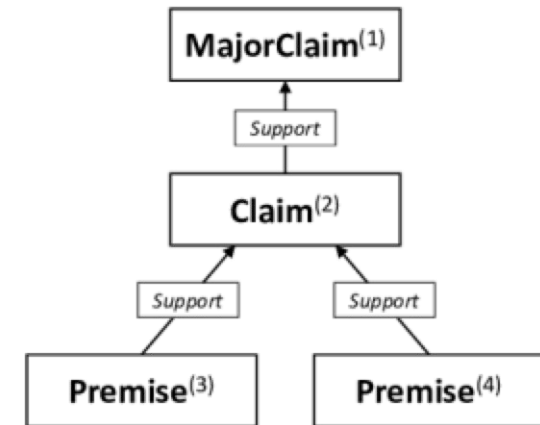
values
ethics
norms



3. Responsibility

Argumentation theory can help!

- the argumentation structure is **encoded in the system**
- argumentation is performed by the system
- the arguments themselves are **provided by humans**
(interactively, text-mining, KGs...)





3. Challenges for a Responsible HI

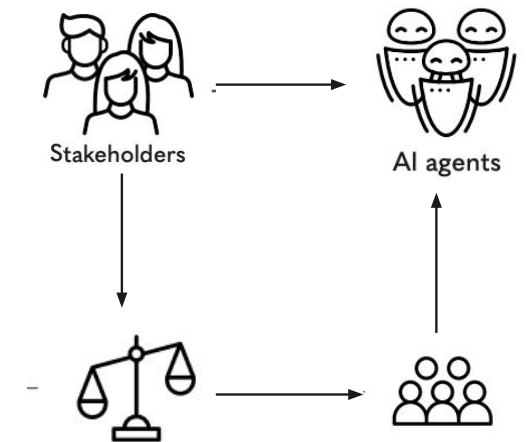
How to account for ethics **during the computation process**?

- **Methods** to identify stakeholders, values and goals, conflicts, align values and goals
- **Encode/model** moral reasoning
- Allow humans to express their norms and values to the system at **runtime** (through human-machine interaction)

3. Examples

Collaborative HI agents **modelling societal values**

- Sociotechnical systems for decision-making (public administration, hospitals, courts...)
- **Prevent** sociotechnical harms (e.g. stereotyping) using situated design (system context)
- Detect stakeholder's **value preference** using NLP + behavioural analysis
- Gathering consensus using social choice techniques





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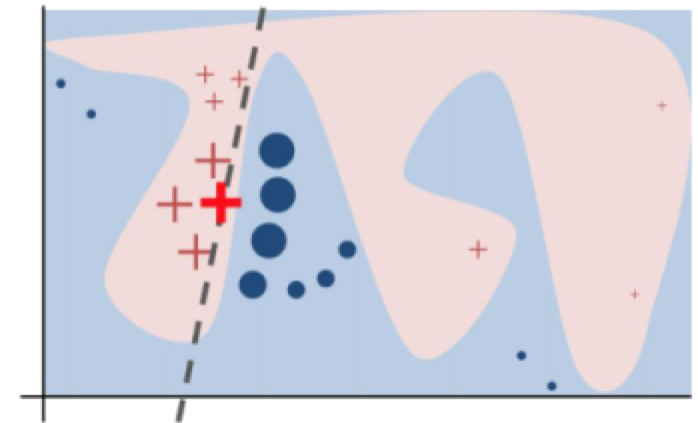
4. Explainability

- Current approaches for explanations are **data-driven**
- Only developers can really understand these explanations
- **Trust from users** is essential in collaboration



(d) Bird - 100%, Person - 39% (e) Importance map of 'bird' (f) Importance map of 'person'

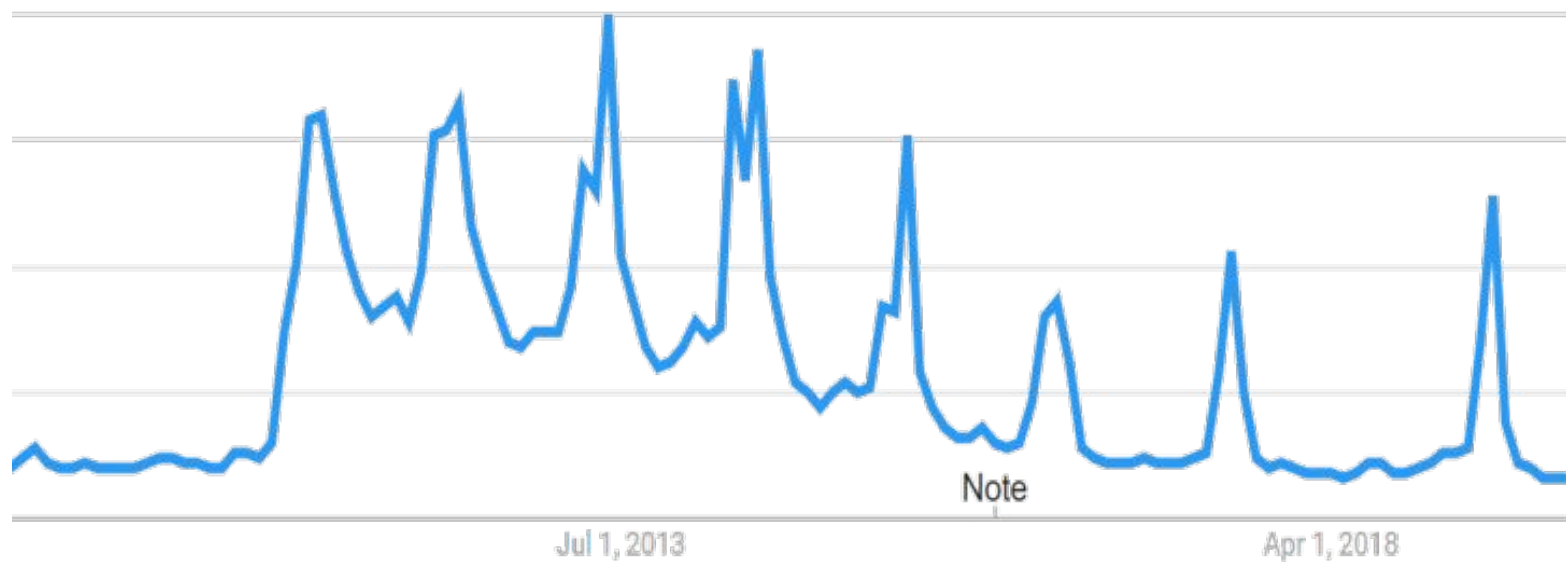
Saliency maps.



Lime/SHAP.

4. Explainability

Humans explain with **background knowledge**



Why do people search for "**a Song of Ice and Fire**" only at specific times of the year?

4. Explainability

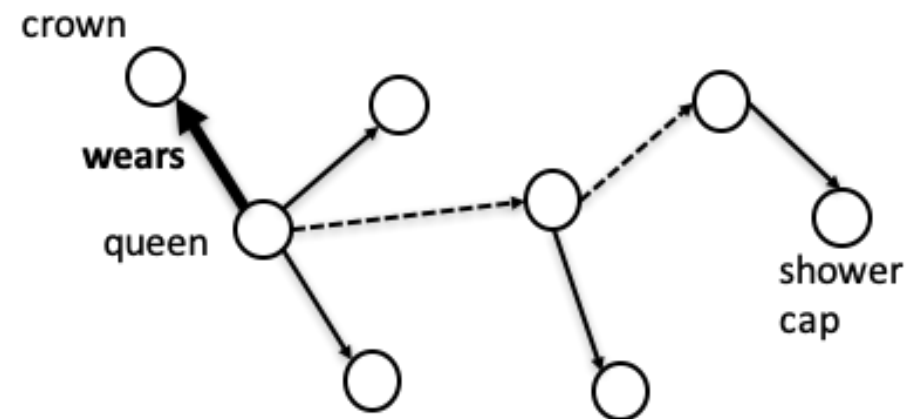
Humans explain with **shared** background knowledge



Class: 793
Label: n04209133
(shower cap)
Certainty: 99.7%

4. Explainability

New approach : **knowledge graphs** as background knowledge to **explain data**





4. Challenges for an Explainable HI

How to build

- **Faithful** explanations (explain the mechanics of the model, with some higher level of abstraction)
- **Rational** reconstructions (give a justification for the decision)
- **Contrastive** explanations (explain why something happened instead of something else)
- **Social** explanations (relate the explanation to the receiver's goals & beliefs)



4. An Example

Dedalo : Explaining with Knowledge Graphs

- Take any **ML output** data



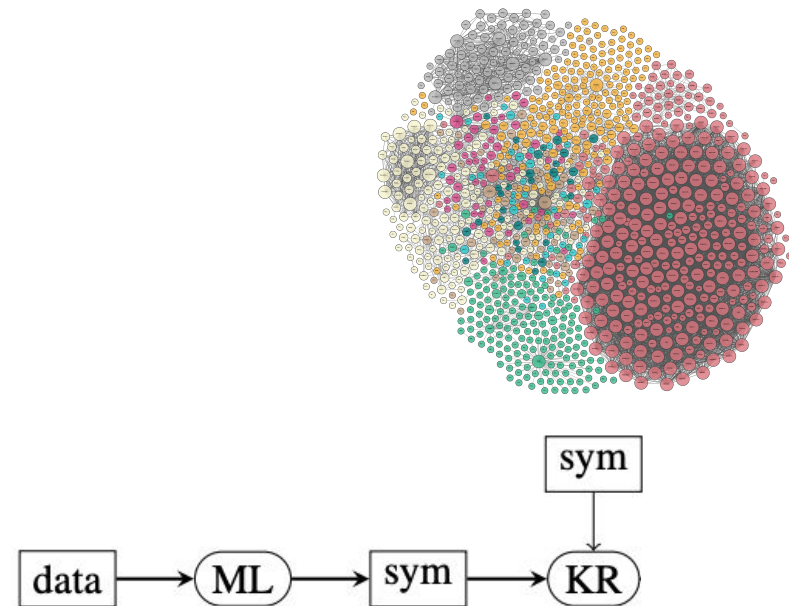
Tiddi, I. et al. (2014). [*Walking Linked Data: a graph traversal approach to explain clusters*](#). COLD @ ISWC 2014.



4. An Example

Dedalo : Explaining with Knowledge Graphs

- Take any **ML output** data
- Use **KR techniques** knowledge about such data (triples in knowledge graphs)



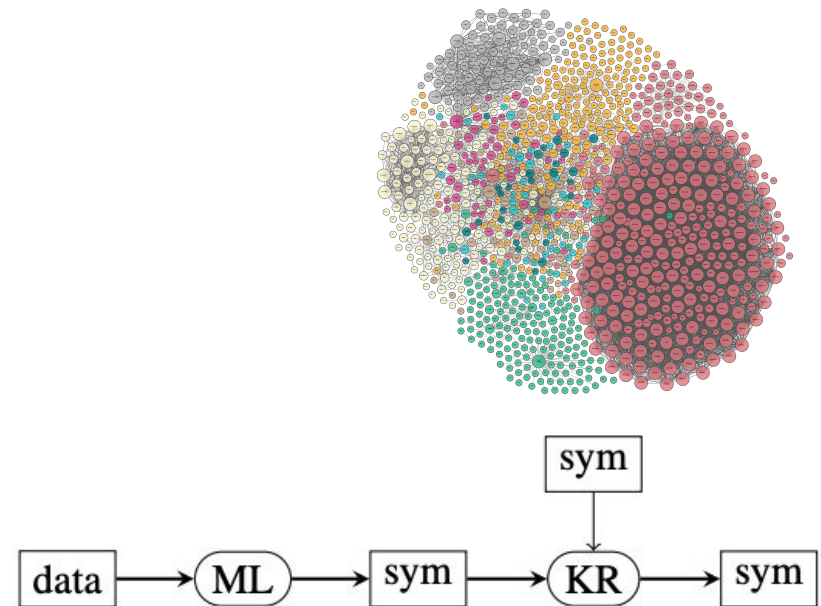
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4. An Example

Dedalo : Explaining with Knowledge Graphs

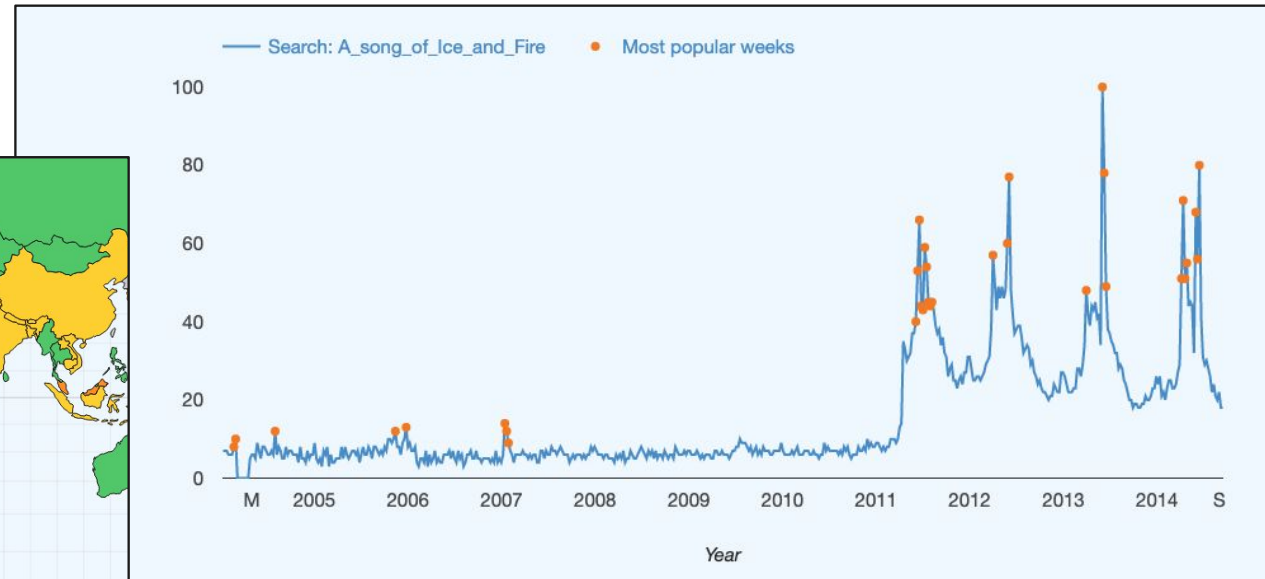
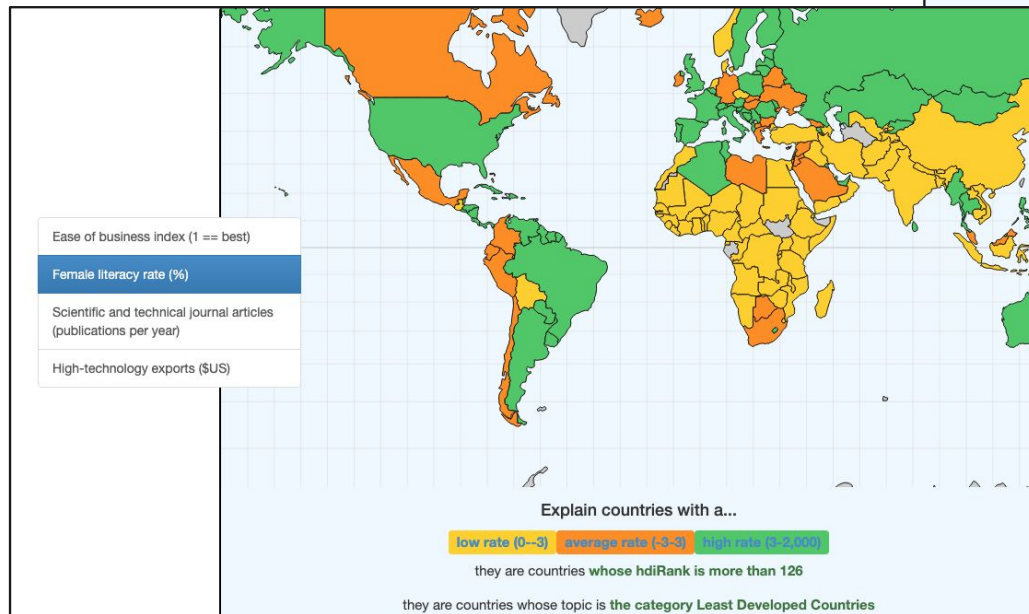
- Take any **ML output** data
- Use **KR techniques** knowledge about such data (triples in knowledge graphs)
- Use the knowledge to **derive explanations** about subsets of data



Tiddi, I. et al. (2014). [Walking Linked Data: a graph traversal approach to explain clusters](#). COLD @ ISWC 2014.



4. An Example



People search for "A song of Ice and Fire" because a new season of the Game of Thrones TV series is out

Countries have a female low literacy rate because they are less developed countries

Demo : <http://dedalo.kmi.open.ac.uk/>



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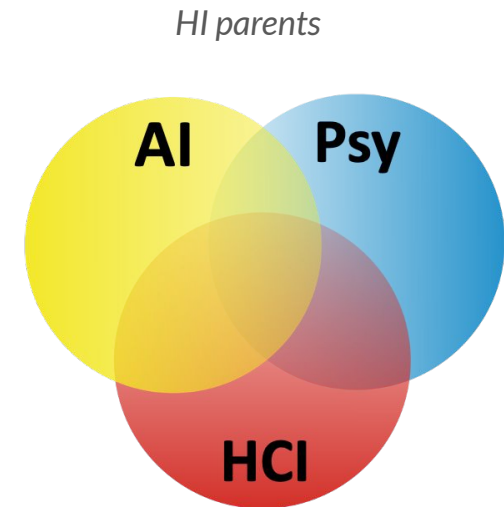
Explainability

...and a **theory** for HI

5. A theory for Hybrid Intelligence

A science needs a theory composed of shared

- assumptions
- language
- questions
- methods
- metrics
- Results



Shall we look at our parents or develop our own?

5. Assumptions for Hybrid Intelligence

From AI

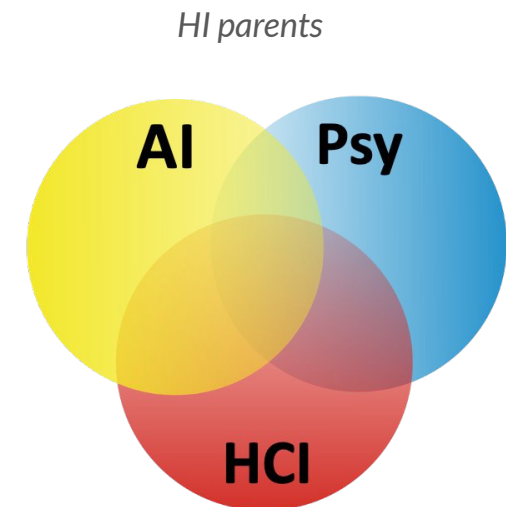
- Intelligence is computational
- Systems have agency
- Individual agents act rationally

From Psychology

- Human behaviour can be predicted
- Humans are information processors (a.o.)

From HCI

- Humans and AI systems have different and complementary strengths and weaknesses
- The relation between humans and machines is **asymmetric**





5. A language for Hybrid Intelligence

- **common knowledge roles**
(ontology)
- **Recurrent tasks**
 - Classical
(recognition, recommendation)
 - HI-specific (creativity, team awareness, multimodality)

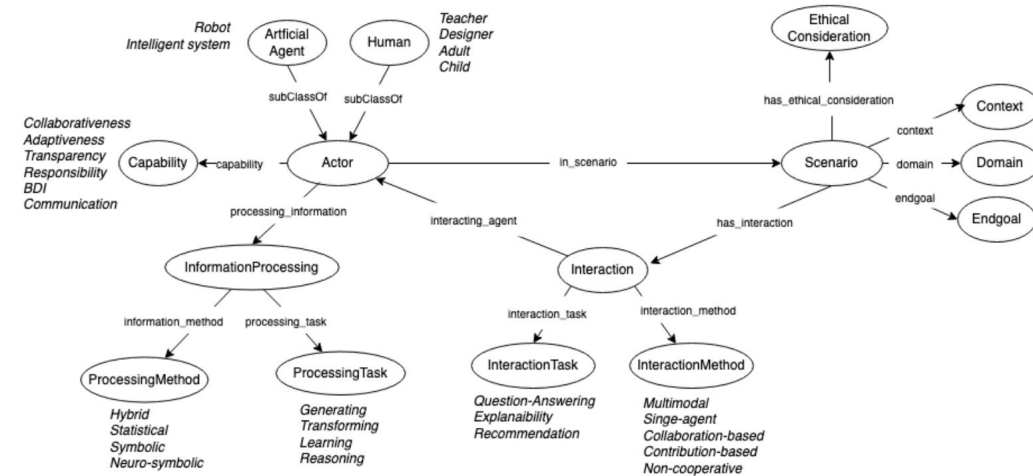


Table 1: Non-exhaustive HI task and subtask decomposition.

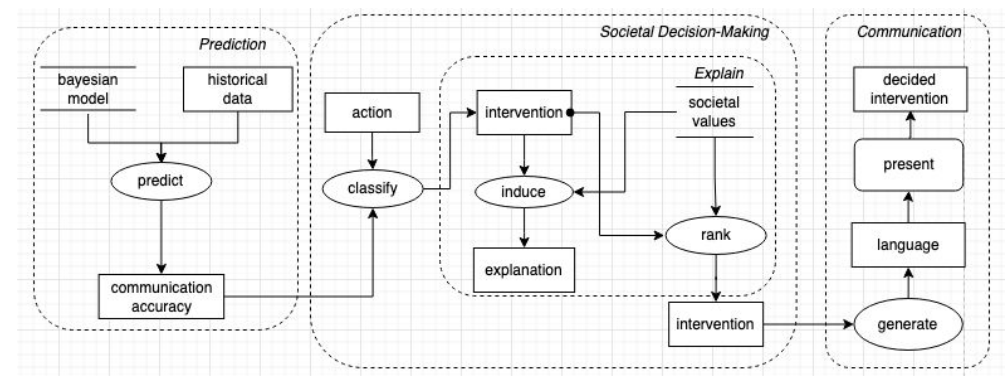
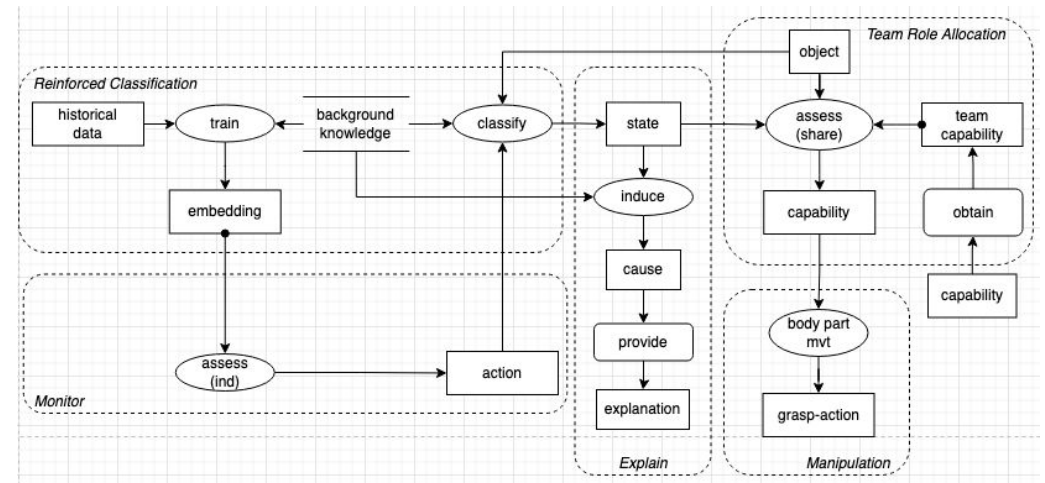
Recognition	Training, Classification		
Prediction	Regression	Training, Predicting	
Reasoning	Inference	Induction, Abduction, Deduction	
Action	Cognitive	Attention	Monitor, Orienting, Sustained
		Memory	Recollect, Short-term, Sensory
		Perception	Vestibular, Olfaction, Audition, Touch, Vision
		Decision Making	Individual, Societal, Debiasing
		Knowledge-aid	Mental Models, Team Role Allocation, Creativity
	Physical	Movement	Body Part Mvt., Translation



5. A language for Hybrid Intelligence

Description and comparison of HI scenarios:

1. A mixed team performing manipulation tasks
2. An artificial agent resolving dilemmas in group decisions (e.g. a jury)



Tiddi et al (2023 under review). *Knowledge Engineering for Hybrid Intelligence*.



A recap of Hybrid Intelligence

Hybrid Intelligence

- AI to **enhance** human capabilities (as others tools do)
- **Collaborate** with humans instead of replacing them

HI functionalities (with **practical examples!**)

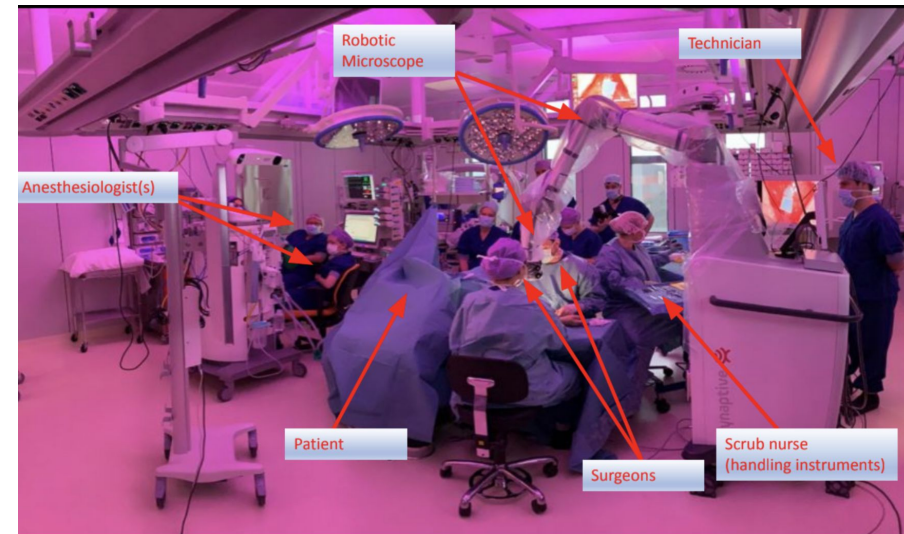
1. Collaboration → from a theory about the world to **the others' theories** of the world
2. Adaptivity → from Machine Learning to **informed Reinforcement Learning**
3. Responsibility → from implicitly to **explicitly coded** norms&values through argumentation
4. Explainability → from data-driven to **hybrid explanations** (data&knowledge-driven)

+ a **theory** for Hybrid Intelligence

HI application : The micro-surgeon

A **coupled system** (human-machine) learning high quality **task performance** in a (micro)surgical environment

- human has an incentive to teach the robot
- the robot learns from the human experience
- the robot asks for questions
- the robot asks for permission
- the robot defers to the human



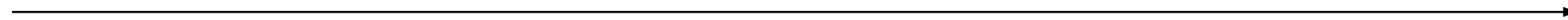
Successful surgical procedure = successful patient outcome



HI application : The micro-surgeon

Low-level

High level



***A. Data
exploration / ML***

***B. Situated
communication
interface***

***C. Collaboration
dynamics***

Tasks

Motion forecasting
Style modeling
Synthetic data generation

Social signal processing, shared
mental models,
Explicit vs. implicit communication

Expectation study
Context-aware collaboration
Role assignment

Techniques

Supervised and unsupervised
learning
Generative models

Data-driven ML
Foundational models (LLMs)
Multimodal ML...
Knowledge Representation
technologies

Design patterns
Wizard of Oz setup...
Simulations

Hybrid Intelligence 2020-2030

<https://www.hybrid-intelligence-centre.nl/>

And do not forget the

[Hybrid Intelligence Conference \(HHA12025\)](#)



Slide credits to the whole HI team





Questions?

All slide credits to Frank van Harmelen and the HI team



Logistics : Hackathon

- Remember to **fill the preference** for the specialties/components
- Be on time (10:00)
- Final best videos will be shown on the social event
- Winner gets a prize !!
- Remember to **enjoy**

Questions?